

Array

Usually if people want to input number to array they will input it started from index 0 until n-1 using for. But, Bibi is bored to code like that she didn't want to input the number that way. So Bibi challenged you to make a program that will read a sequence (represent index) that she made, than input the number to an array but input it with the same sequence as sequence that Bibi gave.

Format Input

The first line represent integer n the size of Bibi's Array. The next line consist n integers A_i representing the sequence that Bibi want, it is guaranteed that the number is distinct. The next line consist n integers representing the value that she want to put inside array with index A_i

Format Output

n integers representing the array that Bibi has starting from index 0.

Constraints

$1 \leq N \leq 1000$

$0 \leq A_i < n$

Sample Input 1	Sample Output 1
5 0 1 2 3 4 1 2 3 4 5	1 2 3 4 5

Sample Input 2	Sample Output 2
5 4 3 2 1 0 1 2 3 4 5	5 4 3 2 1

Sample Input 3	Sample Output 3
5 0 4 3 1 2 1 2 3 4 5	1 4 5 3 2

Notes

There isn't any space after the last number

In the third sample Bibi want to input the number to array index 0 then 4 then 3 then 1 then 2